

Software Requirements Specification

AI Call Agent System

College Administration Outbound Calling Platform

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0 Contents

1 Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the complete functional and non-functional requirements for the AI Call Agent system. The system automates outbound telephone communication between college administration and students/parents using artificial intelligence in English, Hindi, and Gujarati.

1.2 Scope

The AI Call Agent is an outbound-only automated telephony system deployed on a college's on-premises Ubuntu 22.04 server. It enables administrative staff to schedule calls; the system then autonomously dials recipients, conducts spoken conversations using local AI models, and logs results to a database.

In scope:

- Outbound PSTN calls via Plivo SIP gateway
- Multilingual speech recognition (En/Hi/Gu)
- LLM-based natural language response generation
- Multilingual text-to-speech synthesis
- Web UI for call scheduling and monitoring
- Call logging and audio recording storage

Out of scope:

- Inbound call handling
- SMS or WhatsApp messaging
- Mobile app for staff
- Integration with college ERP/LMS systems (future phase)

1.3 Definitions and Acronyms

1.4 References

- IEEE Std 830-1998: IEEE Recommended Practice for SRS
- Plivo Voice API Documentation: plivo.com/docs/voice
- LiveKit Server Documentation: docs.livekit.io
- Qwen3 Model Card: huggingface.co/Qwen/Qwen3-14B
- IndicConformer Paper: AI4Bharat, 2023
- IndicF5 TTS: AI4Bharat, 2024

2 Overall Description

2.1 Product Perspective

The AI Call Agent operates as a standalone system within the college's local network. It interfaces externally only with the Plivo cloud gateway (for PSTN call routing) and optionally

Term	Definition
STT	Speech-to-Text: converts spoken audio to written transcript
LLM	Large Language Model: AI model generating natural language responses
TTS	Text-to-Speech: converts text to synthesized spoken audio
PSTN	Public Switched Telephone Network: global telephone infrastructure
SIP	Session Initiation Protocol: standard for VoIP call signaling
RTP	Real-time Transport Protocol: carries audio/video over IP networks
WebRTC	Web Real-Time Communication: browser-native real-time media protocol
ARI	Asterisk REST Interface: HTTP API for programmatic call control
DID	Direct Inward Dialing: a virtual phone number
PBX	Private Branch Exchange: internal telephone switch/server
Plivo	Cloud SIP gateway service used for PSTN connectivity
LiveKit	Open-source WebRTC server acting as media bridge
Celery	Python distributed task queue for async job processing
Redis	In-memory data store used as Celery message broker
HTMX	HTML extension library for server-driven UI updates

with Groq API and Sarvam AI (cloud fallback for AI inference). All core AI processing occurs on-premises.

Design philosophy: Internet dependency is minimized. Calls and AI processing work entirely on the local network. Internet is required only for Plivo call routing and optional cloud AI fallback.

2.2 Product Functions

1. **Call Scheduling:** Staff schedules outbound calls via web UI with recipient details and call purpose
2. **Automated Dialing:** System places PSTN calls automatically at scheduled time
3. **Speech Recognition:** Transcribes recipient's spoken words in real time (En/Hi/Gu)
4. **Response Generation:** LLM generates contextually appropriate reply
5. **Speech Synthesis:** Converts response text to natural-sounding speech
6. **Conversation Management:** Maintains multi-turn conversation context
7. **Call Logging:** Records call metadata, transcript, and outcome to database
8. **Audio Storage:** Saves call audio recordings to MinIO object store
9. **Dashboard:** Real-time status monitoring and historical call reports

2.3 User Classes

2.4 Constraints

- Server must run Ubuntu 22.04 LTS
- AI models must run locally (data privacy requirement)

User Class	Role	Interaction
Administrative Staff	Call scheduler	Schedules calls, views reports, manages contact lists
Department Head	Supervisor	Views dashboards, approves call campaigns
System Administrator	IT operator	Manages server, updates models, monitors health
Recipient	Student/Parent	Receives and responds to automated calls (voice only)

- System must support English, Hindi, and Gujarati
- Single concurrent call per Plivo DID number (scalable with additional numbers)
- Call audio must not be transmitted to any external server except Plivo for routing
- Must operate on college LAN without internet for core functionality

3 Functional Requirements

3.1 FR-01: User Authentication

hdr	hdr
FR-01.1	System shall provide login page with username and password
FR-01.2	System shall support role-based access (Admin, Staff, Viewer)
FR-01.3	Session shall expire after 8 hours of inactivity
FR-01.4	System shall log all login and logout events with timestamp and IP

3.2 FR-02: Contact Management

hdr	hdr
FR-02.1	System shall allow adding individual contacts (name, phone, type: student/parent)
FR-02.2	System shall support bulk import of contacts via CSV file
FR-02.3	System shall validate phone numbers (10-digit Indian mobile format)
FR-02.4	System shall allow editing and deleting contacts
FR-02.5	System shall support grouping contacts by department, year, or custom tag

hdr	hdr
FR-03.1	System shall allow scheduling a call to a single contact at a specific datetime
FR-03.2	System shall allow scheduling a call campaign to a contact group
FR-03.3	System shall allow selecting call purpose/script from predefined templates
FR-03.4	System shall show preview of selected script before confirming schedule
FR-03.5	System shall allow cancelling a scheduled call before it is placed
FR-03.6	System shall enforce minimum 3-minute gap between consecutive calls (single DID)
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FR-04.1	System shall place outbound PSTN call via Plivo API at scheduled time
FR-04.2	System shall retry call once if not answered (after 5 minutes)
FR-04.3	System shall detect call answer and initiate AI conversation
FR-04.4	System shall detect voicemail/IVR and leave a pre-recorded message
FR-04.5	System shall end call after conversation completion or 5 minutes of silence
FR-04.6	System shall update call status in real time (Scheduled/Dialing/Active/Completed/Failed)
hdr	hdr
FR-05.1	System shall transcribe recipient's speech using IndicConformer model
FR-05.2	System shall support English, Hindi, and Gujarati languages
FR-05.3	System shall detect end-of-utterance using voice activity detection (VAD)
FR-05.4	System shall fall back to Sarvam AI API if local STT fails
FR-05.5	STT shall complete transcription within 800ms of utterance end
hdr	hdr
FR-06.1	System shall generate responses using Qwen3-14B model via Ollama
FR-06.2	LLM shall receive system prompt defining call purpose and context
FR-06.3	LLM shall maintain full conversation history within the call
FR-06.4	LLM shall generate responses in the same language as the recipient's utterance
FR-06.5	System shall fall back to Groq API if local LLM response exceeds 5 seconds
FR-06.6	LLM response shall not exceed 3 sentences for voice delivery

3.3 FR-03: Call Scheduling**3.4 FR-04: Outbound Call Execution****3.5 FR-05: Speech Recognition (STT)****3.6 FR-06: Response Generation (LLM)****3.7 FR-07: Speech Synthesis (TTS)****3.8 FR-08: Call Logging and Storage****3.9 FR-09: Dashboard and Monitoring****4 Non-Functional Requirements**

hdr	hdr	
FR-07.1	System shall synthesize speech using IndicF5 model	
FR-07.2	System shall support English, Hindi, and Gujarati voice output	
FR-07.3	System shall stream synthesized audio to LiveKit as it is generated	
FR-07.4	TTS shall complete synthesis within 500ms for a 2-sentence response	
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FR-08.1	System shall log each call: start time, end time, duration, status, outcome	
FR-08.2	System shall store full conversation transcript linked to call record	
FR-08.3	System shall save call audio recording to MinIO object storage	
FR-08.4	System shall retain call records for minimum 1 year	
FR-08.5	System shall allow staff to view and export call history as CSV	
hdr	hdr	
FR-09.1	Dashboard shall show real-time status of all active and scheduled calls	
FR-09.2	Dashboard shall show daily/weekly/monthly call statistics	
FR-09.3	System shall send alert (email or in-app) if call failure rate exceeds 20%	
FR-09.4	System shall display AI model health status (STT/LLM/TTS operational)	
hdr	hdr	hdr
NFR-P1	End-to-end latency	≤ 4 seconds per conversational turn (local LLM)
NFR-P2	STT latency	≤ 800 ms after end-of-utterance
NFR-P3	LLM latency	≤ 3 seconds (local); ≤ 800 ms (Groq fallback)
NFR-P4	TTS latency	≤ 500 ms for 2-sentence response
NFR-P5	Web UI response	≤ 1 second for page loads
NFR-P6	Concurrent calls	1 per DID; scalable to N with N DID numbers
hdr	hdr	hdr
NFR-R1	System uptime	≥ 99% during college working hours (8am–8pm)
NFR-R2	Call success rate	≥ 95% of scheduled calls attempted
NFR-R3	Data loss	Zero tolerance for call logs; audio recordings best-effort
NFR-R4	Recovery time	System auto-restart within 60 seconds of crash (Docker)

- No mobile app required (responsive web sufficient)
- Served over HTTPS via Nginx on port 443

hdr	hdr
NFR-S1	All web traffic must use HTTPS (TLS 1.2+) via Nginx
NFR-S2	Plivo credentials must be stored as environment variables, never in code
NFR-S3	Database must not be accessible from outside the local network
NFR-S4	Call audio must not be transmitted outside college LAN (except Plivo for routing)
NFR-S5	All API endpoints must require authentication
NFR-S6	Passwords must be hashed using bcrypt (min cost factor 12)
hdr	hdr
NFR-U1	Staff should be able to schedule a call in under 3 clicks
NFR-U2	Web UI must be accessible on Chrome, Firefox, and Edge browsers
NFR-U3	Web UI must be responsive for tablet (1024px+) and desktop screens
NFR-U4	System must display clear error messages in English when a call fails
hdr	hdr
NFR-M1	All services must run in Docker containers for easy replacement
NFR-M2	AI models must be swappable without changing pipeline code
NFR-M3	System must expose health check endpoints for each service
NFR-M4	Configuration must be fully externalized via .env file

5.2 Hardware Interfaces

- College PC: Ubuntu 22.04, NVIDIA GPU (8GB+ VRAM recommended), 16GB+ RAM, 500GB+ SSD
- Network: 100 Mbps LAN; internet required only for Plivo

5.3 Software Interfaces

hdr	hdr	hdr
Plivo API	HTTPS REST	Initiate outbound calls; receive call events
Plivo Webhook	HTTPS POST	Receive call status updates
LiveKit Server	WebRTC/SIP	Media bridge between SIP and AI pipeline
Ollama API	HTTP REST	Run Qwen3-14B inference locally
Groq API	HTTPS REST	Cloud LLM fallback
Sarvam AI API	HTTPS REST	Cloud STT fallback
PostgreSQL	TCP 5432	Primary data store
Redis	TCP 6379	Celery message broker
MinIO	HTTP 9000	Audio file object storage

6 System Constraints and Assumptions

6.1 Constraints

1. **Budget:** Monthly operational cost must not exceed Rs. 2,000
2. **Hardware:** Must run on existing college server hardware
3. **Language:** Must support En/Hi/Gu without fine-tuning
4. **Privacy:** Student call data must remain on college servers
5. **Compliance:** Must comply with TRAI regulations for automated calling (India)

6.2 Assumptions

1. College server has NVIDIA GPU with at least 8GB VRAM
2. Stable internet connection available for Plivo API calls
3. Students are reachable on Indian mobile numbers
4. Call volume does not exceed 100 calls/day in initial deployment
5. College has registered with Plivo and obtained DID number

7 Appendix: Requirement Traceability Matrix

hdr	hdr	hdr	hdr	hdr
FR-01	Authentication	Must Have	1	Planned
FR-02	Contact Management	Must Have	1	Planned
FR-03	Call Scheduling	Must Have	1	Planned
FR-04	Call Execution	Must Have	1	Planned
FR-05	STT (IndicConformer)	Must Have	1	Planned
FR-06	LLM (Qwen3-14B)	Must Have	1	Planned
FR-07	TTS (IndicF5)	Must Have	1	Planned
FR-08	Logging & Storage	Must Have	1	Planned
FR-09	Dashboard	Should Have	2	Planned
NFR-P	Performance	Must Have	1	Planned
NFR-R	Reliability	Must Have	1	Planned
NFR-S	Security	Must Have	1	Planned